



Shivering snakes

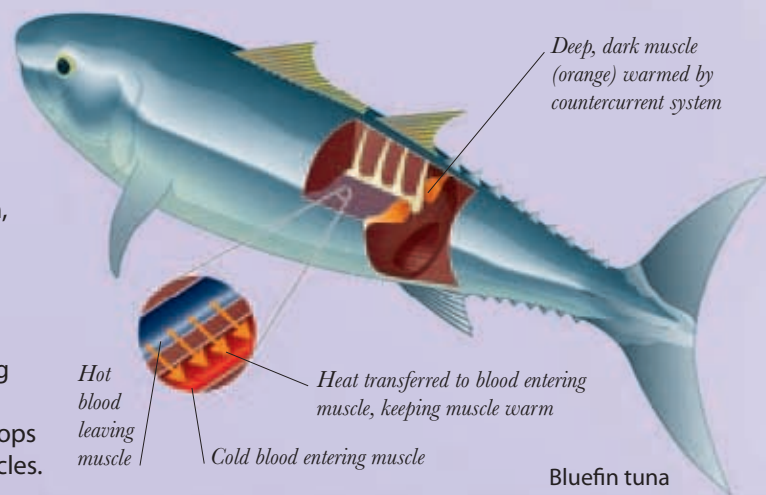
Snakes are cold-blooded, but this diamond python has an unusual way of warming up its eggs. By shivering its muscles, it generates heat, which keeps the eggs up to 9°F (5°C) warmer than the surrounding air. Snakes are not the only cold-blooded animals that shiver. In cold habitats, some moths and bumblebees warm up by shivering before they fly.

Group hug

Watched over by a pair of adults, emperor penguin chicks huddle together on the ice. They are insulated by body fat and by their soft gray down, but huddling gives them an extra way of staying warm. These huddles, or crèches, can contain thousands of young birds. The chicks keep their heads down, out of the wind, and they keep shuffling toward the center of the crèche, so that each one gets its share of the warmth. When the chicks are five months old, they start to grow their adult plumage. The crèches break up, and the young penguins head off for the sea.

Superfast fish

Most fish are cold-blooded, so their temperature is the same as the water around them. Tuna and swordfish are an exception—their central swimming muscles stay warm, letting them speed continuously through the seas. These muscles have a dark color and a special blood supply. Warm blood flowing out of the muscles meets cold blood flowing in, and transfers most of its heat. This system—called a countercurrent—stops warmth from escaping from the muscles.



Bluefin tuna

